

Final Examination Simulation

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Instruction: This exam consists of conceptual questions and computational problems based on advanced asymptotic and discrete methods. Calculation skills are not emphasized; full credits are given for correct principles, steps, and interpretations. One double-sided A4 cheat sheet is allowed.

Useful integral formulas:

$$\int_0^{+\infty} e^{-xu^2} du = \frac{1}{2} \sqrt{\frac{\pi}{x}}, \quad x > 0, \quad \int u dv = uv - \int v du$$

1. **(Concept Question)** (20 points, 4 points each)
 - (a) Give the definition of asymptotic expansion for $f(\varepsilon)$, and briefly describe the main distinction between asymptotic expansion and convergent Taylor series.
 - (b) Briefly tell the differences between regular perturbation and singular perturbation based on the structure of differential equations.
 - (c) Explain why terms like e^{-1/ε^2} will invalidate regular perturbation expansion as $\varepsilon \rightarrow 0$.
 - (d) Briefly compare the applicable integral types of Laplace's method and stationary phase method and their dominant contribution positions.
 - (e) Briefly explain why eigenvalue problems are introduced in the mathematical study of WKB approximation.
2. **(Asymptotic Expansion)** (10 points) Consider the function $f(x) = \sqrt{25+x}$ as $x \rightarrow 0$. We are given a non-power asymptotic sequence base: $\phi_0(x) = 1$, $\phi_1(x) = \sin x$, $\phi_2(x) = \cos x - 1$.
 - (a) Based on the formal definition of an asymptotic sequence, determine and prove whether this given base forms a legitimate asymptotic sequence as $x \rightarrow 0$.
 - (b) Show the exact algebraic limit step to calculate the leading-order coefficient a_0 for the function $f(x)$.
 - (c) Based on the sequential remainder condition, show the exact algebraic limit step to calculate the first-order coefficient a_1 .

3. **(Laplace's Method)** (12 points) Use Laplace's method to find the leading-order asymptotic approximation of the following integral as the parameter $x \rightarrow +\infty$:

$$I(x) = \int_0^1 e^{-x(\cosh t - 1)} dt$$

- (a) Identify the maximum point of the phase function. Explain how the exponential amplification mechanism allows us to compress the global integration interval into a localized domain $[0, \delta]$.
 - (b) Introduce an appropriate rescaled variable to stretch the localized domain, and show the detailed procedure of extending the integration upper limit to positive infinity $(+\infty)$.
 - (c) Substitute the provided integral formula to obtain the final leading-order asymptotic behavior as $x \rightarrow +\infty$.
4. **(Boundary Layer Theory)** (15 points) Consider the following singularly perturbed boundary value problem:

$$\varepsilon y'' + (2 + 3\varepsilon)y' + 6y = 0, \quad y(0) = 2, \quad y(1) = 1, \quad 0 < \varepsilon \ll 1$$

- (a) Without making any asymptotic approximations or dropping the small parameter ε , write down the exact characteristic equation of this ODE, find its exact roots, and write down the exact algebraic solution $y(x)$ of this boundary value problem.
 - (b) Determine the position of the boundary layer (left boundary $x = 0$ or right boundary $x = 1$), and derive the characteristic thickness scale $\delta(\varepsilon)$ of the inner layer using the principle of dominant balance.
 - (c) Write down the simplified characteristic equations for the outer region and the inner region respectively under the asymptotic limit. Find their corresponding asymptotic general solutions.
 - (d) Complete the asymptotic matching using Van Dyke's rule to determine all unknown constants and construct the global composite solution.
5. **(WKB Method and Turning Point)** (12 points) Consider the high-frequency oscillatory linear ODE $\varepsilon^2 y'' - xy = 0$ defined on the domain $x \in [-1, 1]$.
- (a) Identify the turning point of this system. Distinguish the rapid oscillatory solution region and the exponential solution region based on the sign of the coefficient.

- (b) Derive the standard leading-order WKB ansatz for the oscillatory region.
- (c) Introduce a localized rescaled coordinate $\tilde{x} = x/\varepsilon^\alpha$ near the turning point $x_0 = 0$. Use the principle of dominant balance to uniquely derive the inner layer thickness scale index α , and write down the resulting localized inner equation (Airy equation).

6. **(Two-Time Scale Analysis)** (12 points) Consider the weakly nonlinear oscillator equation:

$$y'' + y + \varepsilon(y')^3 = 0, \quad 0 < \varepsilon \ll 1$$

- (a) Introduce the two independent time scales $T_0 = t$ and $T_1 = \varepsilon t$. Rewrite the partial derivative operators for $\frac{d}{dt}$ and $\frac{d^2}{dt^2}$ up to $O(\varepsilon)$.
- (b) Expand the solution as $y = y_0 + \varepsilon y_1 + \dots$. Substitute it into the ODE to obtain the $O(1)$ control equation, and state its general solution via the characteristic equation.
- (c) Apply the solvability condition to eliminate the secular resonance terms, and derive the amplitude evolution equation.

7. **(1D Bifurcation Theory)** (14 points) Given the continuous one-dimensional dynamical system $\dot{x} = \mu - x^2$:

- (a) Calculate all fixed points of the system as the parameter μ varies.
- (b) Analyze the topological stability of each fixed point branch by calculating the linearized eigenvalue.
- (c) Identify the critical bifurcation point μ_c , state the specific name of this bifurcation type, and sketch its standard bifurcation diagram.

8. **(Discretization and Difference Equation)** (5 points) Consider the differential equation

$$y'' + 2\varepsilon y' + y = 0.$$

Let $y_n = y(t_n)$ with constant step size Δt . The first derivative is given below:

$$y'|_n \approx \frac{y_{n+1} - y_{n-1}}{2\Delta t}.$$

Construct the difference approximation for y'' by yourself, substitute all difference expressions into the ODE and derive the corresponding second-order linear difference equation.

期末考试模拟

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考试说明: 本试卷包含概念简答与计算大题, 侧重方法原理理解与步骤推导, 不侧重纯数学计算技巧。允许携带一页 A4 纸正反面手写公式备忘。

可能用到的积分公式:

$$\int_0^{+\infty} e^{-xu^2} du = \frac{1}{2} \sqrt{\frac{\pi}{x}}, \quad x > 0, \quad \int u dv = uv - \int v du$$

1. 概念简答与辨析题 (20 分, 每小题 4 分)

- 简述函数渐近展开的定义, 并简要说明渐近展开与收敛泰勒级数的主要区别。
- 从微分方程结构角度, 简要说明正则摄动与奇异摄动的区别。
- 说明 $\varepsilon \rightarrow 0$ 时, e^{-1/ε^2} 这类超小量会造成正则摄动失效的原因。
- 简要说明拉普拉斯方法与驻相法适用的积分类型, 以及积分主要贡献出现的位置。
- 从数学角度简要说明 WKB 理论中引入特征值问题的缘由。

2. 渐近展开 (10 分) 考虑函数 $f(x) = \sqrt{25+x}$ 在 $x \rightarrow 0$ 时的展开。现给定一组非幂级数的函数序列基底: $\phi_0(x) = 1$, $\phi_1(x) = \sin x$, $\phi_2(x) = \cos x - 1$ 。

- 严格根据渐近序列的严密数学定义, 判定并证明该组基底在 $x \rightarrow 0$ 时是否构成合法的渐近序列。
- 展示通过求解相应的代数极限, 计算出该函数领先阶系数 a_0 的精确步骤;
- 根据渐近展开的逐次余项约束条件, 展示计算第一阶系数 a_1 的精确极限推导步骤。

3. 拉普拉斯积分方法 (12 分) 利用拉普拉斯方法求解大参数积分的领先阶渐近近似, 其中大参数满足 $x \rightarrow +\infty$:

$$I(x) = \int_0^1 e^{-x(\cosh t - 1)} dt$$

- 确定该积分相位函数的极大值点。阐述指数放大效应是如何允许我们将全局积分区间压缩到该点局域邻域 $[0, \delta]$ 内的;
- 引入局域拉伸变量对坐标进行重标度以放大该区域, 并详细展示将积分上限延拓至 $+\infty$ 的完整代数推导过程;

- (c) 代入卷面给出的积分公式，最终求出该积分在 $x \rightarrow +\infty$ 时的领先阶渐近形式。

4. **边界层理论** (15 分) 求解小参数奇异摄动常系数边值问题：

$$\varepsilon y'' + (2 + 3\varepsilon)y' + 6y = 0, \quad y(0) = 2, y(1) = 1, \quad 0 < \varepsilon \ll 1$$

- (a) 不进行任何渐近近似，直接写出该常系数 ODE 的精确特征方程，求出其特征根并写出该边值问题的代数精确解 $y(x)$ 。
- (b) 判断该边值问题边界层发生的位置（左边界 $x = 0$ 还是右边界 $x = 1$ ），并利用主导平衡法推导计算出边界层的特征厚度尺度 $\delta(\varepsilon)$ ；
- (c) 写出在摄动极限下，外层区域与内层区域弱化后的特征方程，并分别求出其渐近通解形式；
- (d) 利用范戴克匹配原理完成内、外解的渐近拼接，确定所有未知常数并写出全局复合解。

5. **WKB 方法与转折点理论** (12 分) 针对定义在区间 $x \in [-1, 1]$ 上的高频摄动线性方程 $\varepsilon^2 y'' - xy = 0$ ：

- (a) 识别该方程的转折点位置。根据对应系数的正负号，划分振荡解区域与指数解区域；
- (b) 推导出振荡区域内的标准领先阶 WKB 渐近表达式；
- (c) 在转折点 $x_0 = 0$ 附近引入坐标 $\tilde{x} = x/\varepsilon^\alpha$ ，利用主导平衡求 α ，并写出内层艾里方程。

6. **双时间尺度分析** (12 分) 求解弱非线性方程：

$$y'' + y + \varepsilon(y')^3 = 0, \quad 0 < \varepsilon \ll 1$$

- (a) 取快慢尺度 $T_0 = t, T_1 = \varepsilon t$ ，将一阶、二阶全导数改写为尺度偏导算子（展开至 $O(\varepsilon)$ ）；
- (b) 做渐近展开 $y = y_0 + \varepsilon y_1 + \cdots$ ，代入方程得到零阶控制方程并写出通解；
- (c) 利用可解性条件消去久期项，推导振幅演化方程。

7. **一维分岔理论** (14 分) 给定系统 $\dot{x} = \mu - x^2$ ：

- (a) 对任意参数 μ 求系统全部不动点；

- (b) 线性化求特征值, 判定各不动点稳定性;
- (c) 求临界分岔参数 μ_c , 注明分岔名称并手绘分岔简图。

8. **离散化与差分方程** (5 分) 已知方程:

$$y'' + 2\varepsilon y' + y = 0$$

记 $y_n = y(t_n)$, 等距步长 Δt , 给出一阶差分形式:

$$y'|_n \approx \frac{y_{n+1} - y_{n-1}}{2\Delta t}.$$

自行构造二阶导数的差分近似, 代入原式整理得到二阶线性差分方程。

Standard Answers and Explanations

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1. Concept Question Answers (20 points)

- (a) Asymptotic expansion: $f - \sum_{k=0}^N a_k \phi_k = o(\phi_N)$ for fixed $N, \varepsilon \rightarrow 0$; Taylor converges as $N \rightarrow \infty$ at fixed variable, asymptotic series may diverge with growing N .
- (b) Regular perturbation: leading coefficient nonvanishing at $\varepsilon \rightarrow 0$; singular: leading term degenerates, boundary layer / WKB appears.
- (c) $e^{-1/\varepsilon^2} \ll \varepsilon^k, \forall k > 0$, all power-series coefficients zero, regular power expansion cannot capture this term.
- (d) Laplace for real exponential phase, localize at global maximum; stationary phase for oscillatory integral, localize at stationary points of phase.
- (e) Homogeneous boundary condition turns free parameter into discrete eigenvalues; λ controls sign of Q , determines oscillatory/exponential; double turning points quantize spectrum via Bohr–Sommerfeld rule.

2. Asymptotic Expansion Answers (10 points)

- (a) $\sin x = o(1), \cos x - 1 = o(\sin x), x \rightarrow 0$, valid asymptotic sequence.
- (b) $a_0 = \lim_{x \rightarrow 0} \sqrt{25 + x} = 5$.
- (c) $a_1 = \frac{1}{10}$.

3. Laplace's Method Answers (12 points)

- (a) Max at $t = 0$, exponential decays rapidly away from $t = 0$.
- (b) $\cosh t - 1 \approx t^2/2$, set $u = t\sqrt{x/2}$, extend upper limit to infinity.
- (c) $I(x) \sim \sqrt{\frac{\pi}{2x}}$.

4. Boundary Layer Theory Answers (15 points)

- (a) $r_1 = -3, r_2 = -2/\varepsilon$.

- (b) Boundary layer at $x = 0$, thickness $\delta = \varepsilon$.
- (c) Outer: $y_{\text{out}} = Ce^{-3x}$; inner: $y_{\text{in}} = C_1 + C_2e^{-2X}$.
- (d) $y_{\text{comp}} = e^{3(1-x)} + (2 - e^3)e^{-2x/\varepsilon}$.

5. WKB Method and Turning Point Answers (12 points)

- (a) TP: $x = 0$; $x < 0$ oscillatory, $x > 0$ exponential.
- (b) Standard WKB oscillatory formula.
- (c) $\alpha = 2/3$, Airy equation $y_{\tilde{x}\tilde{x}} - \tilde{x}y = 0$.

6. Two-Time Scale Analysis Answers (12 points)

- (a) $\frac{d}{dt} = \partial_{T_0} + \varepsilon \partial_{T_1}$, $\frac{d^2}{dt^2} = \partial_{T_0}^2 + 2\varepsilon \partial_{T_0} \partial_{T_1}$.
- (b) $y_{0TT} + y_0 = 0$.
- (c) Solvability removes secular terms and yields amplitude ODE.

7. 1D Bifurcation Theory Answers (14 points)

- (a) $\mu < 0$ no fixed point; $\mu > 0$, $x = \pm\sqrt{\mu}$.
- (b) $+\sqrt{\mu}$ stable, $-\sqrt{\mu}$ unstable.
- (c) Saddle-node bifurcation at $\mu_c = 0$.

8. Discretization and Difference Equation Answers (5 points) $y'' \approx \frac{y_{n+1} - 2y_n + y_{n-1}}{\Delta t^2}$,
substitute and arrange:

$$(1 + \varepsilon \Delta t)y_{n+1} + (\Delta t^2 - 2)y_n + (1 - \varepsilon \Delta t)y_{n-1} = 0.$$